GEOMETRY QUEST · WORLD 16

Fractals

WORKSHEET

Name _____ Date _____ Stars I give myself: ☆ ☆ ☆

1 Cantor set: each step removes the middle third of every piece. Complete the table.

step	pieces	length left (start = 1)
0	1	1
1	2	2/3
2		
3		
4		

2 Koch snowflake: start with a triangle of side 9 cm. Each step, every edge becomes 4 edges, each a third as long. Complete the table.

step	edges	perimeter
0	3	27 cm
1		
2		
3		

3 Draw step 1 of the Koch curve on this 9 cm line: split it into three 3 cm parts, draw an equilateral triangle on the middle part, then rub out its base.

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4 Fractal hunt: find three things in nature or at home that repeat their pattern when you zoom in. Draw one of them.

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0